DURAIRAJAN G

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PROFESSIONAL SUMMARY

Final-year B.Tech student in Artificial Intelligence and Data Science with hands-on machine learning, deep learning, and generative AI experience: multi-agent analytics platforms, offline retrieval pipelines, fine-tuned language models, and CNNs. Skilled at turning data and language models into reliable, deployable solutions.

EDUCATION

K.S. Rangasamy College of Technology | B.Tech, AI and Data Science | 2023 – Present | CGPA: 8.41/10
Coursework: Machine Learning, Data Mining, DBMS, Software Engineering, Agentic AI

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java, C, R
- **Frontend:** React, Vite, Tailwind CSS, Framer Motion
- **Generative AI:** RAG, LLM Fine-Tuning, Prompt Engineering, Hugging Face, Ollama, GGUF Quantization, LangChain, LangGraph
- **Agentic Systems:** Autonomous Agents, Multi-Agent Orchestration, Tool Integration, Workflow Automation
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, Keras, Neural Networks, CNNs, Text Classification, Feature Engineering, Model Evaluation
- **Retrieval & NLP:** FAISS, Vector Databases, Embeddings, Semantic Search, PyMuPDF
- **Deployment:** FastAPI, REST APIs, Docker, Docker Compose, GitHub Actions, Jenkins, CI/CD, Streamlit, Git
- **Data & BI:** Pandas, NumPy, SciPy, Plotly, Matplotlib, Jupyter, EDA, Power BI, Tableau
- **Databases:** MySQL, PostgreSQL, SQLite, MongoDB, Redis, SQLAlchemy

PROJECTS

AI Data Analyst Platform | *Python, FastAPI, LangGraph, Pandas, Redis, MySQL, React* | GitHub
- Architected a LangGraph multi-agent system that profiles CSV/Excel data, plans analysis, validates results, and reports findings in plain English.
- Built 15+ deterministic Pandas tools, Plotly charts, and a FastAPI service routing between Gemini and Groq, with optional Tavily research.
- Added Redis caching, JWT authentication, MySQL storage, Docker Compose deployment, GitHub Actions CI, and 72 passing tests.

Offline RAG Document Assistant | *Python, FAISS, Ollama, LangChain* | GitHub
- Engineered an offline PDF Q&A assistant using retrieval-augmented generation and an open-source LLM.
- Indexed PyMuPDF-parsed chunks as local embeddings for semantic lookup.
- Cited file name, page number, similarity score, and confidence with every answer.

AI Ticket Routing System | *Llama 3.2 3B, Hugging Face, GGUF, Streamlit* | GitHub
- Designed a classifier that routes queries to Sales, Support, or Technical teams.
- Fine-tuned the base network on a cleaned, labeled JSONL corpus.
- Compressed the result into GGUF for a cached, instant-response web interface.

AI Digit Recognition System | *TensorFlow, Keras, FastAPI, React, Tailwind CSS* | GitHub
- Trained a CNN on the Kaggle MNIST dataset and served predictions through a FastAPI endpoint.
- Built an animated dashboard with a live drawing canvas, confusion matrix, accuracy/loss charts, and prediction history.

WORK EXPERIENCE

Data Analytics & ML Intern | Imagivite Technology Pvt. Ltd., Tiruchengode, India | Jul 2025 – Aug 2025
- Explored, cleaned, and selected features from raw datasets with Pandas and Scikit-learn.
- Delivered full machine learning pipelines spanning training, validation, and metric evaluation.
- Translated findings into business reports backed by charts and predictive forecasts.

ACHIEVEMENTS & CERTIFICATIONS

- **Digital Engineering Certification** – NASSCOM/ER&D, Gold grade, 74% score.
- **1st Prize, Short Film Contest** – VYuga Ability Carnival, Nexyuga Innovations (LiCRA Cultural Club team).
- **Campus Ambassador Certificate** – RINEX Pvt. Ltd., for outstanding initiatives (Oct–Nov 2023).